

LIVE EVENT ORCHESTRATION

FULL TESTS PASSED: 13/13

VERSION 1.0.0

StagePilot (StageFlow) – Master Project Overview

AI Co-Pilot & Real-Time Synchronization Platform for Live Events & Stage Anchors

1. Problem & Mission Statement

During live tech summits and hackathons, organizers backstage and anchors on stage frequently fall out of sync when sessions overrun. Traditional cue sheets fail instantly when delays occur, international speaker names get mispronounced, and stage anchors are left improvising. **StagePilot** eliminates this friction through real-time WebSocket communication, intelligent cascading schedule delays, and context-aware AI script assistance.

2. System Architecture

Organizer War-Room (Port 5173)

Real-time agenda visualizer, 1-click cascading delay buttons (+5m, +10m, +15m), and instant flash emergency alert broadcaster.

Stage Anchor Station (Port 5173)

High-contrast teleprompter with glass mirror mode (scaleX(-1)), speaker phonetic guides, synthesized sound alerts, and live countdown timer.

Node.js + Express API (Port 5000)

Modular REST controllers, JWT & Bcrypt authentication, input validation, and dynamic prompt assembly.

WebSocket Engine & DB

Socket.IO bi-directional rooms ('event_id'), MongoDB with automatic in-memory fallback for zero-config startup.

3. Key Innovations

Feature	Capability & Implementation
Cascading Delay Engine	Injecting +5m or +10m automatically recalculates the start and end times for all downstream agenda sessions without page reload.
Teleprompter Glass Mirroring	Horizontal mirroring toggle ('scaleX(-1)') designed specifically for physical beamsplitter teleprompter glass rigs.
Phonetic Pronunciation	International speaker names display phonetic spelling guides (e.g., <i>Dr. Elena Rostova</i> → [eh-LEH-nah ross-TOH-vah]).
Zero-Refresh Emergency Chimes	Synthesizes a pure 800Hz → 600Hz two-tone audio alert using the browser's Web Audio API (zero audio file dependencies).

Feature	Capability & Implementation
Context-Aware AI Co-Pilot	Backstage drawer powered by Gemini/OpenAI SDKs with an offline-resilient smart fallback engine for stage banter and delay scripts.

4. Quick Start & Execution

```
# 1. Start Express + Socket.IO Backend Server (Port 5000) cd server npm start # 2.
Start React + Vite Frontend Client (Port 5173) cd client npm run dev
```

5. 1-Click Demo Accounts

Role	Email	Password	Dashboard Route
Organizer	organizer@stagepilot.io	password123	http://localhost:5173/organizer
Stage Anchor	anchor@stagepilot.io	password123	http://localhost:5173/anchor